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Downhole perforating gun tube and components

Abstract

A gun tube for a downhole perforating gun assembly includes a body having a cavity and one or more weights in the cavity. Gravity acts on the weights, which causes the tube body to rotate around its longitudinal axis so the weights are adjacent the lower part of the wellbore in which the gun assembly is positioned when oriented horizontally. This position shape charges in the gun tube in a desired position prior to the shape charges being fired. The gun tube may also include one or more end connectors with electrical contacts having a first, extended position, and a second, contracted position, and/or end connectors that can be assembled by hand without the use of tools.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of co-pending U.S. patent application Ser. No. 18/751,039 filed Jun. 21, 2024, which is a continuation of U.S. patent application Ser. No. 18/656,426 filed May 6, 2024, which is a continuation of U.S.

patent application Ser. No. 18/110,804 filed Feb. 16, 2023-now U.S. Pat. No. 11,976,539, which is a continuation of U.S. patent application Ser. No. 17/380,490 filed Jul. 20, 2021-now U.S. Pat. No. 11,624,266, which is a continuation of U.S. patent application Ser. No. 16/293,508 filed Mar. 5, 2019-now U.S. Pat. No. 11,078,762, each of which is herein incorporated by reference in its entirety.

FIELD OF THE INVENTION

(1) The present invention relates to components for perforating wellbores.

BACKGROUND

(2) When drilling oil or gas wells, a wellbore is formed. After drilling, the drill string and bit are removed, and the remaining wellbore is lined with a metal casing. A generally annular area is formed between the outside surface of the metal casing and the surrounding formations.

(3) A cementing operation is typically conducted to fill the area between the metal casing and the surrounding formation with concrete. The combination of concrete and metal casing strengthens the wellbore.

(4) Later, perforations are usually made in the metal casing and concrete using a perforating gun assembly that is generally comprised of a steel carrier, and a charge tube inside of the carrier with shaped charges positioned in the charge tube. The perforating gun is lowered into the wellbore and is typically connected to an electric wireline or other conveyance device until it is at a predetermined position. Then a signal actuates a firing head of the gun, which detonates the shaped charges in the gun. The explosion of the shaped charges perforates the metal casing and concrete to allow fluids to flow from the formation into the wellbore.

SUMMARY

(5) The present disclosure includes for perforating gun tubes (also referred to herein as “gun tubes,” “tubes,” “guns,” or “charge tubes”) and related structures and components. In one embodiment, a gun tube may include a body, one or more weights in a cavity of the body, and one or more end fittings. Gravity acts on the weights, which causes the gun tube to rotate around its longitudinal axis when the gun is horizontally oriented so the one or more weights are adjacent the bottom of the wellbore. The explosive charges (also called “shape charges”), which are in the gun tube, then point upwards and/or downwards, or in any direction dictated by the position of the one or more weights. The gun tube may include one or more end fittings that include a bearing housing and permit the gun tube body to rotate relative to the end fittings. The gun tube may include tabs that retain the one or more weights in the cavity. There may be multiple sets of tabs so the weights can be positioned and retained at different locations in the cavity in order to position the explosive charges at a desired location relative the one or more weights.

(6) Alternatively, the gun could be rotated by a motor in accordance with a signal generated by a human or machine operator. A sensor could be on the gun, or on a carrier that positions the gun in the wellbore. The sensor would detect the position of the gun and of shape charges in the gun tube relative the wellbore and transmit a signal, or cause a signal to be transmitted, that includes the gun tube's rotational position in the wellbore. An operator could then signal the motor to rotate the gun until the shape charges are at a desired position before the shape charges are fired.

(7) In another embodiment, the one or more weights in the cavity are connected to a rotatable plate at one or both ends of the gun tube. For example, if there are two weights, one would be inside the cavity and attached to a first rotatable plate at a first end of the gun tube. The other weight could be attached to a second rotatable plate at the second end of the gun tube. In this embodiment, the weights are not fixed in the cavity, and as the plates rotate, the weights rotate inside of the cavity. When the plates are fixed in position, such as with fixation pins, the weights are fixed in position in the cavity. The position of the weights in the cavity determines the firing direction of the explosive charges when the gun tube is in a horizontal position in a wellbore.

(8) A gun tube according to this disclosure could also include one or two end fittings that include

end connectors. Each end connector has an electrical contact that is biased to a first, extended position, and that can be moved to a second, compressed position when compressive axial force is applied to the electrical contact.

(9) The end connectors may also be configured to attach to the end fitting without tools. An end connector may be inserted into a support of the end fitting by hand and then rotated and released to be retained in the support. Disassembly, if desired, is also done by hand. The end connector would be pressed inward relative the support, and rotated to a position at which it would be released and then separate from the support.

(10) A dual plunger may be utilized as an electrical connection through a sub-assembly used with one or two gun tubes. The dual plunger has at least a first conductive stem, which is preferably biased to a first, extended position, and preferably also has a second conductive stem, which is preferably biased to a first, extended position. Each stem may be moved to a second, compressed position when compressive axial force is applied to the end of the stem. The first conductive stem and second conductive stem can move independently of each other. The plunger could have one end formed to be rotated by a tool in order to be threaded into a sub-assembly. For example, an end of the plunger may have a hexagonal shape.

(11) Because the plungers are removable, and thereby interchangeable, the conductive stems can be designed or configured for any form of electrical contact required.

(12) A double wire through with ground connector (“DWG”) could be used instead of a dual plunger in a sub-assembly to transmit electricity to fire the shape charges in a gun tube. If a DWG is used end connectors are not required in the end fittings of the gun tube because electricity could be transferred from wires connected to the DWG directly to the shape charges. Alternatively, end connectors could still be used.

(13) A DWG includes a first conductive stem that may or may not have a first, extended position and a second, compressed position, in the same manner as a conductive stem of the plunger. The DWG also preferably has one or more exterior grounding arms to securely ground to an inner bore of a sub-assembly when the DWG is positioned in the central bore of the sub-assembly. An insulative, protective sheath, which could be wire harness assembly, can be positioned on a second stem of the DWG for the secure connection of wires.

(14) A rubber or plastic (such as silicone rubber) dart retainer may be used with a dual plunger or DWG in place of a metal retainer where a grounding connection or secure method of constraining the dual plunger or DWG is not required. The dart retainer helps to insulate the sub-assembly to prevent shorts, by preventing loose wires from contacting the sub-assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective, side view of a gun tube in accordance with aspects of the invention.
- (2) FIG. 2 is an exploded, perspective side view of a first end cap of the gun tube of FIG. 1.
- (3) FIG. 3 is an exploded, perspective side view of a second end cap of the gun tube of FIG. 1.
- (4) FIG. 4 is a partially exploded, perspective side view of the gun tube of FIG. 1.
- (5) FIG. 5 is a side view of the gun tube of FIG. 1.
- (6) FIG. 5A is an end view of the gun tube of FIG. 4.
- (7) FIG. 5B is an opposite end view of the gun tube of FIG. 4.
- (8) FIG. 6 is a cross-sectional side view of the gun tube of FIG. 4 taken along line A-A of FIG. 5A.
- (9) FIG. 7 is a cross-sectional top view of the gun tube of FIG. 4 taken along line B-B of FIG. 5A.
- (10) FIG. 8 is a cross-sectional top view of the gun tube of FIG. 4 taken along line C-C of FIG. 5B.
- (11) FIG. 9 is a break-away, side perspective view of the gun tube of FIG. 1.
- (12) FIG. 10 is a close-up, side, perspective view showing detail D of FIG. 9.

- (13) FIG. 11 is a close-up, side, perspective view showing detail E of FIG. 9.
- (14) FIG. 12 is a side, perspective view of an end connector in accordance with an embodiment of the invention.
- (15) FIG. 12A is a side view of the end connector of FIG. 12.
- (16) FIG. 12B is a cross-sectional, side view of the end connector of FIG. 12A.
- (17) FIG. 12C is an end view of the end connector of FIG. 12A.
- (18) FIG. 12D is an alternate, side view of the end connector of FIG. 12.
- (19) FIG. 12E is a rotated, alternate side view of end connector of FIGS. 12 and 12D.
- (20) FIG. 12F is a rotated, alternate side view of end connector of FIGS. 12 and 12D.
- (21) FIG. 12G is a perspective, front end connector view of FIG. 12.
- (22) FIG. 12H is an end view of the end connector of FIG. 12E.
- (23) FIG. 12I is an alternate, side perspective view of the end connector of FIG. 12.
- (24) FIG. 13 is a partial, cut-away, perspective view of a support and a side, perspective view of an end connector.
- (25) FIG. 13A is a partial, cut-away, perspective view of a support with the end connector of FIG. 13.
- (26) FIG. 13B is an alternate, cut-away, perspective view of a support with the end connector of FIG. 13.
- (27) FIG. 13C is an alternate, cut-away, perspective view of a support with the end connector of FIG. 13.
- (28) FIG. 13D is a partial, cut-away, side perspective view of a support and a side, perspective view of an end contact.
- (29) FIG. 13E is an alternate, cut-away, side perspective view of a support and a side, perspective view of an end contact.
- (30) FIG. 13F is an alternate, cut-away, side perspective view of a support and a side, perspective view of an end contact.
- (31) FIG. 13G is an alternate, cut-away, side perspective view of a support and a side, perspective view of an end contact.
- (32) FIG. 13H is a side, perspective view of a support and end connector.
- (33) FIG. 13I is a side, perspective view of a support and end connector assembled.
- (34) FIG. 13J is a cross-sectional, side perspective view of the support and end connector of FIG. 13I.
- (35) FIG. 14 is a side, perspective view of a plunger.
- (36) FIG. 14A is a side, perspective, cross-sectional view of the plunger of FIG. 14.
- (37) FIG. 14B is a side view of the plunger of FIG. 14.
- (38) FIG. 14C is an end view of the plunger of FIG. 14.
- (39) FIG. 14D is an alternate end view of the plunger of FIG. 14.
- (40) FIG. 14E is a perspective, side view of the plunger of FIG. 14.
- (41) FIG. 14F is a perspective, end view of the plunger of FIG. 14.
- (42) FIG. 14G is an opposite, perspective, end view of the plunger of FIG. 14.
- (43) FIG. 14H is a perspective, end view of the plunger of FIG. 14.
- (44) FIG. 15 is a side, perspective view of an alternate plunger.
- (45) FIG. 15A is a side, cross-sectional view of the plunger of FIG. 15.
- (46) FIG. 16 is an exploded, perspective view of the plunger of FIG. 14 and a sub-assembly.
- (47) FIG. 16A is an exploded, cross-sectional view of the plunger and a sub-assembly of FIG. 16.
- (48) FIG. 17 is a side view of a sub-assembly with a plunger and small dart retainer.
- (49) FIG. 17A is an end view of the sub-assembly of FIG. 17.
- (50) FIG. 17B is a side, perspective view of the sub-assembly of FIG. 17.
- (51) FIG. 17C is a side, cross-sectional view of the sub-assembly of FIG. 17.
- (52) FIG. 17D is a side, perspective view of the sub-assembly of FIG. 17.

(53) FIG. 17E is a side, perspective, cross-sectional view of the sub-assembly of FIG. 17.

(54) FIG. 18 is a side view of a sub-assembly with a plunger and large dart retainer.

(55) FIG. 18A is an end view of the sub-assembly of FIG. 18.

(56) FIG. 18B is a side, perspective view of the sub-assembly of FIG. 18.

(57) FIG. 18C is a side, cross-sectional view of the sub-assembly of FIG. 18.

(58) FIG. 18D is a perspective, side view of the sub-assembly of FIG. 18.

(59) FIG. 18E is a perspective, side, cross-sectional view of the sub-assembly of FIG. 18.

(60) FIG. 19 is a perspective, side view of a double wire feed through with ground.

(61) FIG. 20 is a side, perspective, cross-sectional view of the double wire feed through with ground of FIG. 19.

(62) FIG. 20A is a top, perspective, cross-sectional view of the double wire feed through with ground of FIG. 19.

(63) FIG. 21 is a side view of the double wire feed through with ground of FIG. 19.

(64) FIG. 21A is an alternate side view of the double wire feed through with ground of FIG. 18.

(65) FIG. 21B is an end view of the double wire feed through with ground of FIG. 21A.

(66) FIG. 21C is an alternate view of the double wire feed through with ground of FIG. 21A.

(67) FIG. 21D is a side, perspective view of the double wire feed through with ground of FIG. 21.

(68) FIG. 21E is an alternate view of the double wire feed through with ground of FIG. 21.

(69) FIG. 21F is a perspective, side view of the double wire feed through with ground of FIG. 21.

(70) FIG. 22 is an end view of an alternate double wire feed through with ground.

(71) FIG. 22A is a cross-sectional side view of the double wire feed through with ground of FIG. 22 taken through line A-A.

(72) FIG. 22B is a bottom view of the double wire feed through with ground of FIG. 22 taken through line B-B.

(73) FIG. 22C is an exploded, perspective view of the double wire feed through with ground of FIG. 22.

(74) FIG. 22D is a perspective, cross-sectional side view of the double wire feed through with ground of FIG. 22.

(75) FIG. 22E is a side, perspective view of the double wire feed through with ground of FIG. 22.

(76) FIG. 22F is a close-up, partial cross-section view of the double wire feed through with ground of FIG. 22 with wires attached.

(77) FIG. 22G is a partial, cross-sectional side view of the double feed through with ground of FIG. 22F positioned in a sub-assembly.

(78) FIG. 23 is an exploded, side perspective view of a gun assembly including an outer casing and two sub-assemblies.

(79) FIG. 24 is a cross-sectional, side, perspective view of the gun assembly of FIG. 23.

(80) FIG. 25 is a side, perspective, assembled view of the gun assembly of FIG. 23.

(81) FIG. 26 is a cross-sectional, side, perspective view of the gun assembly of FIG. 25.

(82) FIG. 27 is a perspective, side view of an alternate gun tube in accordance with aspects of the invention.

(83) FIG. 28 is a perspective, partially-exploded side view of an alternate gun tube in accordance with aspects of the invention.

(84) FIG. 28A is an exploded, perspective side view of a first end cap of the gun tube of FIG. 28.

(85) FIG. 28B is an exploded, perspective side view of a second end cap of the gun tube of FIG. 28.

(86) FIG. 29 is a side view of the gun tube of FIG. 28.

(87) FIG. 29A is an end view of the gun tube of FIG. 29.

(88) FIG. 29B is an opposite end view of the gun tube of FIG. 29.

(89) FIG. 30 is a cross-sectional side view of the gun tube of FIG. 29 taken along line 30-30 of FIG. 29A.

(90) FIG. 31 is a cross-sectional top view of the gun tube of FIG. 29 taken along line 31-31 of FIG. 29A.

(91) FIG. 32 is a cross-sectional top view of the gun tube of FIG. 29 taken along line 32-32 of FIG. 29B.

(92) FIG. 33 is a break-away, side perspective view of the gun tube of FIG. 28.

(93) FIG. 34 is a close-up, side, perspective view showing detail D of FIG. 33.

(94) FIG. 35 is a close-up, side, perspective view showing detail E of FIG. 33.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(95) Turning now to the drawings, where the purpose is to describe embodiments of this disclosure and not to limit the claims, FIGS. 1-13J show a gun tube 10.

(96) Gun Tube

(97) Gun tube 10 has a tube body 12, a first end 14 with a first end fitting 16, and a second end 18 with a second end fitting 20. Gun tube 10 further includes a cavity 114, charge openings 116, charge clip openings 117, and tabs 130. Gun tube 10 is preferably cylindrical and formed of steel.

(98) Charge openings 116 are configured to retain shape (or explosive) charges 122, best seen in FIGS. 1-8. Charge openings 116 can be of a suitable shape, size, and position to hold a specific type or size of shape charge 122, and point the shape charge 122 outward in a specific direction. Charge clip openings 117 are configured so that clips 132 can be positioned on the outer wall of tube body 12. Clips 132 are attached to wires that connect to the shape charges 122 in a manner known to those skilled in the art.

(99) One or more weights 124 are positioned in cavity 114. As shown, there are two weights 124A, 124B, although only one, or more than two, weights may be used. One or more weights 124 can be of any size, shape or weight suitable to move gun tube 10 so that the one or more weights 124 cause gun tube 10 to rotate relative to bearing assemblies 26 so the portion of gun tube 10 that retains one or more weights 124 is at the bottom of the wellbore (i.e., closest to the Earth's center) when gun tube 10 is positioned horizontally in a wellbore. Bearing assemblies 26 allow gun tube 10 to rotate around axis A in either direction relative the first end fitting 16 and the second end fitting 20.

(100) Weight 124A as shown is semi-circular, comprised of steel, fills about half of the volume of cavity 114, in which it is positioned, is juxtaposed first end 14 of gun tube 10 and extends about $\frac{1}{3}$ of the length of gun tube 10. Weight 124A preferably weighs about 1 $\frac{3}{4}$ lbs. at sea level in this embodiment. Weight 124B as shown is semi-circular, comprised of steel, fills about half the volume of cavity 114, in which it is positioned, is juxtaposed second end 18 and extends about $\frac{1}{5}$ of the length of gun tube 10. Weight 124B most preferably weighs about 0.8 lbs. at sea level in this embodiment. The size, weight, and configuration of one or more weights 124 can be varied to any suitable amount depending upon the application and diameter or length of gun tube 10.

(101) Gun tube 10 also includes tabs 130 that are used to retain the one or more weights 124 in cavity 114. In the embodiment shown weight 124A and weight 124B are positioned in cavity 114. Then tabs 130 are pressed down against the flat surface of weight 124A to retain weight 124 in cavity 114, and pressed down against the flat surface of weight 124B to retain weight 124B in cavity 114. Thus, the tabs 130 in the Figures are shown in their pressed down position.

(102) Alternatively, one or more weights 124 may be positioned differently relative to shape charges 122 in gun tube 10 than as shown. When positioned as shown, shape charges 122 will basically face straight upwards and straight downwards when gun tube 10 is positioned horizontally in a wellbore, because gravity pulls the one or more weights 124 to the bottom of the wellbore. If an operator instead wanted the shape charges 122 to be positioned and fired outward at an angle, such as 45°, 60°, or 90°, from straight up or straight down, the one or more weights 124 could be positioned differently in the cavity 114. Then, when gravity pulls and orients the one or more weights 124 to the bottom of the horizontal wellbore, the shape charges 122 would be oriented to fire in the desired direction. So, gun tube 10 can have a plurality of tabs 130 sufficient

to position the one or more weights **124** at multiple locations within cavity **114**. An operator can then select the desired location for the one or more weights within cavity **114** depending on the direction the operator would like shape charges **122** to fire.

(103) End Fittings and End Contacts

(104) First end fitting **16** includes an end contact **22**, an outer collar **24**, a bearing assembly **26**, and a support **28**. Second end fitting **20** has the same structure and components as first end fitting **16**. Second end fitting **20** includes an end contact **22**, an outer collar **24**, a bearing assembly **26**, and a support **28**. Because the respective components of each end fitting **16** and **20** have the same structure, only the components of first end fitting **16** will be described in detail. The same components or structures on second end fitting **20** are designated by the same reference numerals as those for first end fitting **16**.

(105) End contact **22** has a body **42** with a first end **44**, second end **46**, and an annular center **48**. First end **44** has an electrical contact **50**. A stem **52** extends from second end **46**. Stem **52** has an opening **55** to which a wire can be connected. End contact **22** has an internal structure, known to those in the art, that enables electricity to be transmitted from electrical contact **50** to stem **52**, at which point electricity is transferred to one or more wires in electrical communication with stem **52**.

(106) Body **42** is preferably comprised of an insulating material, such as plastic. One or more frangible elements, which are shown, are two tabs, **54** extend outward from second end **46**. As shown, the tabs are rounded and extend outward a maximum of about $\frac{1}{8}$ " to $\frac{5}{16}$ ", or about $\frac{1}{8}$ " to $\frac{1}{4}$ ", or about $\frac{1}{8}$ " to $\frac{3}{16}$ ", or about $\frac{3}{16}$ " to $\frac{1}{4}$ " from body **42**. Another structure, such as a continuous or discontinuous annular ridge, or different shaped structures, could be used as the one or more frangible elements. The tabs are about .080" to .150", or about .10" or about .110", or about .120" thick. Body **42** has a first annular portion **48A**, a second annular portion **48B**, and a central annular position **48C**. A spring **56** is positioned on first annular portion **48A** between central annular portion **48C** and tabs **54**.

(107) The spring **56** used for each end contact **22** can be selected by an operator to be, for example, a high-tension spring, medium-tension spring, low-tension spring, or a spring of any suitable tension for the given application. The spring is selected in a manner known to those in the art, so that it ensures electrical connectivity to a device that electrical contact **50** touches in order to transmit electricity from the device to electrical contact **50**. In one embodiment, electrical contact **50** touches the stem of a plunger, which is described below. In another embodiment, the electrical contact **50** touches a mechanical switch (not shown), which is known to those skilled in the art. The spring pressure exerted by spring **56** must be firm enough to bias electrical contact **50** outward to ensure electrical conductivity, but not so firm that it could prematurely begin setting a mechanical switch due to wellbore vibrations or concussive blasts in adjacent guns.

(108) For example, a spring could be selected to have a compression force of any suitable amount between about 2 lbs. and 10 lbs., or about 3 lbs. to 8 lbs., or about 4 lbs. to 7 lbs., or about 4 lbs. to 6 lbs., or about 5 lbs., or any amount from about 2 lbs. to about 15 lbs., or about 5 lbs. to about 15 lbs.

(109) One or more frangible elements, which as shown are two tabs **54** are breakable (or frangible) from body **42** upon the application of an outward force along longitudinal axis A generated by an explosion of shape charges **122**. One or more frangible elements **54** could break, for example, upon the application of an explosive outward force of: about 30 lbs. or more, about 40 lbs. or more, about 50 lbs. or more, about 60 lbs. or more, about 70 lbs. or more, about 80 lbs. or more, about 90 lbs. or more, about 100 lbs. or more, or any explosive, outward force from about 30-200 lbs. or more, along axis A. The purpose of one or more frangible elements **54** breaking is so the electrical connection to gun tube **10** is broken when the shape charges **122** are exploded. Any suitable structure on end contact **22** could be used for this purpose.

(110) Outer collar **24** is preferably comprised of metal, such as aluminum. Outer collar **24** has a

first end **58**, a second end **60** having an opening **61** and an inner bearing surface **63**, an annular side wall **62**, an opening **64** in first end **58**, a cavity **66**, and one or more openings **68** in side wall **62**. Openings **68** are configured to receive grounding hardware items (such as ball plungers, or a spring and electrically conductive ball staked in place) **70**, or hardware, such as fastener **103**, attaching a ground wire **101**.

(111) Bearing assembly **26** comprises a housing preferably circular in shape and has a first end **72**, a second end **74**, a body **76** with an outer wall **78** and an inner wall **80**, an opening **82** at first end **72**, and opening **83** at second end **74**, and a cavity **84** that retains ball bearings **26A**. Bearing assembly **26** could instead be what persons skilled in the art refer to as a thrust bearing. Any suitable structure to allow the rotation of tube body **12** around axis A may be utilized.

(112) Support **28** is preferably comprised of metal, such as aluminum, and has a first end **86**, a second end **88**, a first body portion **90** that has a top surface **92** and an annular outer wall **94**, a second body portion **96** that has a top surface **98**, and an annular outer wall **100**, and an opening **102** therethrough. Opening **102** has two wing sections **102A** and **102B** sized and shaped so frangible elements (shown here as tabs) **54** of end contact **22** can pass therethrough. Top surface **98** has two wing recesses **103A**, **103B** that are positioned approximately 90° relative wing sections **102A**, **102B**, wherein the recesses **103A**, **103B** are configured to receive and retain one or more frangible elements **54** after they pass through wing sections **102A**, **102B** and end contact **22** is rotated, as described further below. A rib **107** is formed in opening **102**, preferably adjacent recesses **103A**, **103B**.

(113) First end fitting **16** is assembled by placing spring **56** onto first annular portion **48A** of end contact **22** between one or more frangible elements **54** and central annular portion **48C**. Then end contact **22** is pressed through opening **102** of support **28** from second end **88**, as best seen in FIGS. **13-13J**. The one or more frangible elements **54** are aligned with and pushed through wing sections **102A**, **102B** and end contact **22** is then rotated (preferably about) 90° so the one or more frangible elements **54** align with wing recesses **103A**, **103B**. Pressure is released by the assembler and the one or more frangible elements **54** are then received and retained in wing recesses **103A**, **103B**, and end contact **22** is thus connected to support **28** without the use of tools or fasteners.

(114) When tabs **54** are pressed through wing sections **102A**, **102B**, first end **56A** (adjacent one or more frangible elements **54**) of spring **56** presses against rib **107** inside opening **102** of support **28**. When the one or more frangible elements **54** are retained in wing recesses **103A**, **103B**, spring **56** is retained between rib **107** and central annular portion **48C**. Outward pressure (i.e., towards second end **88** and towards first end **14** of gun tube **10**) is applied by spring **56** to end contact **22**, which biases end contact **22** and electrical contact **50** to the first, extended position.

(115) Bearing assembly **26** is positioned over second body portion **96** so that second end **74** and opening **83** are juxtaposed top surface **92** of first body portion **90**.

(116) Outer collar **24** is positioned over end contact **22**, bearing assembly **26** and support **28**, so that electrical contact **50** extends through opening **61** of outer collar **24**, most preferably by any amount from about 1/16" to about 5/16". First end **72** and opening **82** of bearing assembly **26** are then juxtaposed inner bearing surface **63** of outer collar **24**.

(117) One or more grounding hardware items **70** are positioned in one or more openings **68** and are preferably press fit into place and staked. The hardware items **70** are preferably either a ball plunger unit, or a combination of spring and electrically conductive ball bearing staked in place.

(118) A ground wire **101** is connected to support **28** by a screw **103** being passed through lead **101A** and being threaded into opening **29**. An electrical lead **105** may then be positioned over stem **52** by pressing it on where it remains because of a pressure fit electrical lead **105** is preferably comprised of a flexible material such as elastomer. Electrical lead **105** is attached to one or more wires to receive electricity passing through end contact **22**. An advantage of electrical lead **105**, which is an insulative protective sheath with wires already attached, is ease and speed of use, and creating a reliable connection. Presently, wires are placed by hand through opening **55** of stem **52**

and then wrapped around stem **52**, and have a silicone tubing sleeve manually placed over the wire wrapping to provide electrical insulation and to keep stem **52** electrically isolated from the gun tube body **12**.

(119) End contact **22** has a first position at which spring **56** biases it away from second end **88** of support **28**, and outward from first end fitting **16**, as shown, e.g., in FIGS. **1-8**. End contact **22** has a second, contracted position at which spring **56** is fully compressed. The distance between the first position and the second position is at least .150", or at least $\frac{3}{8}$ ", or at least $\frac{1}{2}$ ", or at least $\frac{5}{8}$ ", or at least $\frac{3}{4}$ " or at least 1", or any amount from .150" to 1", or from .150" to 1.250". Known end caps do not compress, or may compress only slightly (e.g., about $\frac{1}{8}$ " or less). The advantage of the outward biasing and travel of the end contact **22** and electrical contact **50** is better reliability in maintaining an electrical connection. When a string of gun tubes **10** are placed in a wellbore as part of an assembly including sub-assemblies **200** (discussed below), stresses on the assembly can create gaps between gun tubes **10** and sub-assemblies **200**. Further, stresses, including downstream shape charges exploding, can cause upstream contacts to press against one another, which can lead to breakage and a gap where there is no electrical contact, or broken components that will no longer function. The outward bias and compressibility of the end contacts **22** help alleviate these problems.

(120) Gun Tube With Indexing Weights

(121) In an alternate embodiment shown in FIGS. **28-35**, a plate **600** or similar structure may be used to index one or more weights **124'** to different positions in cavity **114'** of gun tube **10'**. This allows an operator the flexibility to move one or more weights to a desired location, and when gravity acts upon the weights they are moved to be juxtaposed the bottom of the wellbore in which the gun tube **10'** is positioned. The end fittings **16'** and **20'** in this embodiment again have a rotatable portion that enables the gun tube **10'** to rotate around its longitudinal axis **A'** so that shape charges **122** are oriented properly.

(122) In this embodiment, gun tube **10'** is in all respects the same as gun tube **10** except as described herein and as shown in the figures. Pins **602** and indexing apertures **125**, **125A** retain weights **124A'**, **124B'** in position, as explained below. Gun tube **10'** preferably does not include tabs, such as tabs **130** in gun tube **10**. Optionally, tabs **130** could be utilized to help retain weights **124A'** and **124B'** in position in the manner previously described.

(123) As shown, each weight **124A'** and **124B'** in this embodiment have a semi-cylindrical, concave center portion **124I'**, although each may be of any suitable size, material, and configuration. Each weight **124A'** and **124B'** has a first end **126** 'having a plurality of indexing apertures **125A**. Weight **124A'** as shown has a semi-circular outer surface, is comprised of steel, fills about half of the volume of cavity **114'**, in which it is positioned, is juxtaposed first end **14'** of gun tube **10'** and extends about $\frac{1}{3}$ of the length of gun tube **10'**. Weight **124A'** preferably weighs about 1 $\frac{3}{4}$ lbs. at sea level in this embodiment. Weight **124B'** as shown has a semi-circular outer surface, is comprised of steel, fills about half the volume of cavity **114'**, in which it is positioned, is juxtaposed second end **18'** and extends about $\frac{1}{5}$ of the length of gun tube **10'**. Weight **124B'** most preferably weighs about 0.8 lbs. at sea level in this embodiment. The size, weight, and configuration of one or more weights **124'** can be varied to any suitable amount depending upon the application and diameter or length of gun tube **10'**.

(124) Each of one or more plates **600** is preferably comprised of steel about $\frac{1}{4}$ " to $\frac{1}{2}$ " thick, preferably circular, and has a diameter slightly less than the inner diameter of tube body **12'**. Plate **600** is connected to the wall of cavity **114** (i.e., the inner wall of tube body **12'**) by any suitable means, such as soldering or mechanical fastening. If, for example, weights **124A'**, **124B'** were utilized, one of the plates **600** would be juxtaposed weight **124A** at first end **14'** of tube body **12'** and another plate **600** would be juxtaposed weight **124B** at second end **18'** of tube body **12'**. An operator could then rotate each of the weights **124A'**, **124B'** to a desired location in cavity **114** depending on the direction the operator would like the shape charges **122** to fire, and retain the one

or more weights **124'** in the desired location using a pin **602**.

(125) In this example, utilizing two weights, the weights **124A'**, **124B'** would be moved by rotating each to the same relative position in cavity **114'** and then using a pin **602** to fit through openings **24P'** in each end fitting **16'** and **20'**, through an indexing aperture **125** of each plate **600**, and into an aligned indexing aperture **125A** in weight **124A'** and **124B'**. This retains each weight **124A'**, **124B'** at the desired position in cavity **114'** of gun tube **10'**.

(126) If two plates are used, each plate **600** preferably has the same number of indexing apertures **125** at the same relative locations as the other plate **600**. The indexing apertures **125** preferably include indicia visible on the inner surface **601** (i.e., the surface facing away from an end **14'** or **18'** of gun tube **10'** and towards its center) to identify each indexing aperture **125**, so the same indexed position for each plate **600** could be readily identified by an operator using the indicia. For example, each plate **600** may have eight indexing apertures **125** equally, radially spaced about all or part of the outer portion of the plate **600** (although a plate **600** may include any suitable number of apertures at any suitable locations). To make sure weights **124A'**, **124B'** are the same relative positions in cavity **114'**, the respective apertures on each plate **600** would have the same indicia to designate indexing apertures **125** at the same relative position in cavity **114'**.

(127) For example, if each plate **600** had eight apertures, the apertures could be designated by numerals 1-8. In this example, each weight **124A'**, **124B'** would be at the same radial position in cavity **114'** if a pin **602** was positioned in an indexing aperture **125** designated by the same indicia (such as numeral "4") on each plate **600**. The indexing apertures **125A** in each weight **124A'**, **124B'**, could also include indicia. For example, if each weight **124'** has eight indexing apertures **125A**, these apertures could also be designated by numerals 1-8. Using that example, an operator would know that weights **124A'**, **124B'** would be the same relative position in cavity **114'** if the indexing aperture **125A** designated by the same indicia (such as numeral "4") for each weight **124A'**, **124B'** was aligned with the indexing aperture **125** designated the same indicia (such as numeral "3") in each plate **600**. A pin **602** would then be positioned through opening **24P'** in each end fitting **16'** and **20'**, through the indexing aperture **125** designated as "3" in each plate **600**, and into the indexing aperture **125A** designated as "4" in each weight **124A'** and **124B'**.

(128) First end fitting **16'** is the same as first end fitting **16** except as described here and shown in the figures. Second end fitting **20'** is the same as second end fitting **20** except as described here and shown in the drawings.

(129) Bearing assembly **26'** comprises a housing is preferably circular in shape and has a first end **72'**, a second end **74'**, a body **76'** with an outer wall **78'** and an inner wall **80'**, an opening **82'** and a cavity **84'** that retains ball bearings **26A**. Bearing assembly **26'** could instead be what persons skilled in the art refer to as a thrust bearing. Any suitable structure to allow the rotation of tube body **12** around axis **A'** may be utilized. Bearing assembly **26'** has a smaller diameter than previously described bearing assembly **26** in order to provide space for pin **602**.

(130) Support **28'** is preferably comprised of metal, such as aluminum, and has a first end **86'**, a second end **88'**, a body portion **90'** that has a front surface **92'**, an annular outer wall **94'**, and an opening **102'** therethrough. Part of opening **24P'** is formed through support **28'**. Opening **102'** has two wing sections that are the same as previously described wing sections **102A** and **102B**. The wing sections are sized and shaped so frangible elements (shown here as tabs) **54** of end contact **22** can pass therethrough. Support **28'** fits inside of bearing assembly **26'** and rotates inside of outer collar **24**.

(131) An opening **24P'** is formed in the various components of end fitting **16'** and/or **20'** to permit insertion of a pin **602** through the end fitting **16'** and/or **20'**, through an indexing aperture **125** in a plate **600**, and into an indexing aperture **125A** of a weight **124'**.

(132) Sub-Assembly and Plunger

(133) FIGS. **16-18E** show a sub-assembly **200** having a first end **202** with outer threads **202A** and opening **202B**, a second end **204** with outer threads **204A** and opening **204B**, a central portion **206**,

and a central bore **208** with a first threaded end **208A**, and a second end **208B**. Central bore **208** extends through sub-assembly **200** from opening **202B** to opening **204B**.

(134) The sub-assembly **200** is known in the art and is used to connect two gun tubes **10**, as generally shown in FIGS. **23-26**. Also known in the art is outer casing **700**, usually comprised of steel, that fits over each gun tube **10**. An outer casing protects gun tube **10** as it is moved into and through a wellbore. Each outer casing **700** has a first end **702** with internal threads **702A**, a second end **704** with internal threads **704A**, and a bore **708** extending therethrough. Each of the ends **702**, **704** threadingly connects to an outwardly-threaded end **202** or **204** of a respective sub-assembly **200**, as generally shown in FIGS. **23-26**. In this manner, a string of connected gun tubes **10** is produced.

(135) Sub-assembly **200** requires a device to provide an electrical connection through it from one gun tube **10** to another gun tube **10**. One such a device is referred to herein as a plunger. In FIGS. **14-14H** a plunger **300** is shown. In use, plunger **300** is received in central bore **208** of sub-assembly **200** as shown in FIGS. **16-18E**. Plunger **300** has an outer casing **302** preferably made of insulating material, the outer casing **302** having a first end **301** and a second end **303**, an electrically conductive core **304** with a first stop **306** and a second stop **308**, a first conductive stem structure **310** with a first stem **310A** and a first cylinder **310B** that has a diameter greater than the diameter of the first stem **310A**, a second conductive stem structure **312** with a second stem **312A** and a second cylinder **312B** that has a diameter greater than the diameter of the second stem **312A**, preferably a first spring or other biasing structure **314** between first conductive stem structure **310** and first stop **306**, and a second spring or other biasing structure **316** between second conductive stem structure **312** and second stop **308**. First stem **310A** has a first distal tip **311** and second stem **312A** has a second distal tip **313**. Electrically-conductive core **304** has a first cavity **309** in which spring **314** is positioned and a second cavity **309A** in which spring **316** is positioned.

(136) Outer casing **302** as shown has an annular outer surface with one or more (and as shown, two) annular grooves **315** juxtaposed first end **301**. Each groove **315** includes an O-ring **318**. O-rings **318** can be selected of varying durometers or materials for the environment in which they are used. O-rings **318** create an interference fit in central bore **208** to prevent wellbore liquid from entering central bore **208**. Outer casing **302** at first end **301** has a greater diameter than the rest of outer casing **302**. The increased diameter is any amount from about 100" to .300", and the purpose is to create a snug fit in central bore **208**.

(137) As shown, plunger **300** has two stem structures **310**, **312** that are moveable between a first, extended, position and a second, contracted position, but plunger **300** (or plunger **300'**) could have only one such structure and the other could stem structure could have just one position.

(138) Springs **314**, **316** each permit from about .150" to about 1.250" of travel along longitudinal axis B, of respectively, first conductive stem structure **310** and second conductive stem structure **312**. As shown, each stem structure **310**, **312** has a first, extended position (shown in the figures), and a second, compressed position in which respective springs **314**, **316** are compressed. Each stem structure **310**, **312** can move independently of the other. Springs **314**, **316** can be selected by an operator to have a compressive force suitable for the particular condition to which plunger **300** will be subjected. For example, a spring **314**, **316** may have any compressive force or spring rate between about 2 lbs. and about 40 lbs., such as about 2 lbs. to about 40 lbs., about 2 lbs. to about 15 lbs., about 2 lbs. to about 10 lbs., about 4 lbs. to about 15 lbs., or about 4 lbs. to about 10 lbs., or any force from about 10 lbs. to about 50 lbs., such as about 15 lbs., about 20 lbs., about 25 lbs., about 30 lbs., about 35 lbs., about 40 lbs., about 45 lbs., or about 50 lbs.

(139) The purpose of biasing, moveable stem structures **310**, **312** outward, and to permit their travel along axis B between a first, extended position and a second, compressed position, is to help ensure that an electrical connection is maintained when a string of gun assemblies **10** and sub-assemblies **200** are positioned in a wellbore. The string can be subject to stresses that push the respective components together, which can damage electrical connections if they cannot compress,

and thus can move the respective electrical connections apart. The biasing of the stems outward to an extended position, and the ability of the stems to compress without breaking, helps to alleviate this problem. This structure permits play between the electrical connections, as opposed to a rigid connection that can more easily be damaged.

(140) Plunger **300** could also include exterior grounding arms having the same configuration as exterior grounding arms **414** for DWG **400**, which are shown in the Figures and described below.

(141) Alternately, a plunger **300'**, as shown in FIGS. **15-15A** may be utilized. Plunger **300'** is in all respects the same as plunger **300** except that outer casing **302'** has a uniform outer diameter, so the portion of outer casing **302'** juxtaposed first end **301'** would have the same diameter as the portion juxtaposed second end **303'**.

(142) A metal retainer nut **220** may be screwed into central bore **208** to retain plunger **300** or **300'**, as shown in FIGS. **16, 16A**, which helps retain plunger **300** in central bore **208**. Retainer nut **220** has a central opening **222** in which first stem **310A** is positioned.

(143) Dart Retainer

(144) Each end **202, 204**, or only one end **202** or **204**, of a sub-assembly **200** may include a dart retainer **250** or **380**. Further, a dart retainer **250** or **380** may be used with a double wire through with ground, which is described below. If a dart retainer is used, it would be in place of a metal retainer nut **220**.

(145) As shown in FIGS. **17-17E**, a small dart retainer **250** is an insulating sheath that is preferably comprised of rubber or elastomer, such as silicone rubber. It helps prevent short circuits by a loose wire touching sub-assembly **200**. Only one dart retainer **250** shall be described because if a sub-assembly **200** utilizes two, the second dart retainer **250** would be utilized in the same manner, but would be at second end **208B** of sub-assembly **200** with second stem **312A**.

(146) Dart retainer **250** has a first portion **250B** with a first diameter, a second portion **250A** with a second diameter, and an opening **252** therethrough. Dart retainer **250** is preferably configured so first portion **250B** fits in first threaded end **208A** of central bore **208** and opening **252** at least partially surrounds first stem **310A** of plunger **300**.

(147) Alternatively, as shown in FIGS. **18-18E**, a large dart retainer **380** is an insulating sheath that is preferably comprised of rubber or elastomer, such as silicone rubber. It helps prevent short circuits by a loose wire touching sub-assembly **200**, and also helps prevent shrapnel from damaging the surface of central bore **208**. Only one dart retainer **380** shall be described because if a sub-assembly **200** utilizes two, the second dart retainer **380** would be utilized in the same manner, but would be at second end **208B** of sub-assembly **200** with second stem **312A**.

(148) Dart retainer **380** has a first portion **380B** with the same first diameter as first portion **250B**, a larger second portion **380A** with a diameter greater than that of second portion **250A**, and an opening **382**. First portion **380B** is configured to be positioned in first threaded end **208A** of central bore **208** and opening **382** at least partially surrounds first stem **310A** of plunger **300**. Second portion **380A** is sized to fit against the wall of opening **202B** in order to provide protection and help prevent shorts.

(149) Double Wire Feed Through with Ground

(150) FIGS. **19-22G** show a double wire with ground ("DWG") **400** and **500**. The DWG **400** could be used instead of a dual plunger in a sub-assembly **200** to transmit electricity to a gun tube **10**.

(151) If a DWG is used, end contacts **22** are not required in the end fittings **16, 20** of gun tube **10** because electricity is conducted through wires that are in contact with second conductive stem **412** and with the shape charges **122**. Alternatively, a DWG could be used with an end contact **22**.

(152) DWG **400** is configured to be received in central bore **208** of sub-assembly **200**. DWG **400** has an outer housing **402** preferably made of insulating material, an electrically conductive core **404**, a first end **406**, a second end **408**, a first conductive stem **410**, a second conductive stem **412**, and optionally a spring or other biasing structure between first conductive stem **410** and electrically conductive core **404**.

(153) DWG **400** also preferably has one or more exterior grounding arms **414** to securely ground to the central bore **208** of the sub-assembly **200**. An insulative protective sheath, which may be heat shrink tubing **450**, can be manually placed or affixed over second conductive stem **412** of the DWG **400** for secure attachment of wires **452**, instead of having to connect wires to second conductive stem **412**.

(154) One or more annular grooves **416** are preferably formed on the outer surface of outer housing **402**. Each groove preferably receives an O-ring (or gasket) of varying durometer **418** that pressure fits into central bore **208** of sub-assembly **200**.

(155) One or more exterior grounding arms **414** are positioned adjacent grooves **414A** on outer housing **402**. When DWG **400** is pressed into central bore **208** of sub-assembly **200**, one or more exterior grounding arms **414** press against the annular wall of central bore **208** to help ensure the grounding of DWG **400**.

(156) Intelligent Gun Tube

(157) As shown in FIG. **27**, gun tube **10'** is a smart assembly that is the same in all respects as gun tube **10** except it does not require one or more weights **124** (although it may still include them), and it includes a motor **M** on first end **14** and/or on second end **18**. A motor **M** may be attached to end fitting **16** and/or **20**. An accelerometer or other sensor (e.g., 3-axis (magnetometer), 6 axis (magnetometer plus accelerometer) or **9** (magnetometer plus accelerometer plus gyroscope), degree of freedom ("DOF") device may be used to detect the relative rotational position of gun tube **10'** in a wellbore. The sensor can thus assist an operator in determining the position of the shape charges **122** in the wellbore. The operator can then control the one or more motors to rotate gun tube **10'** and position the shape charges **122** where the operator wants them before firing them. A signal could be sent wirelessly, or by a wired connection, from the sensor to the operator who can use a controller (such as a computer or cell phone) to directly or indirectly operate the one or more motors to orient the gun tube **10'**.

(158) Perforating Gun Assembly

(159) FIGS. **23-26** show a perforating gun assembly **1000**. Gun assembly **1000** includes previously-described gun tube **10**, a previously-described sub-assembly **200**, each of which include a plunger **300**. Alternatively, one or both sub-assemblies could include a previously-described DWG **400** at respective ends **204** of each sub-assembly **200**. In that case, end contacts **22** need not be used. Wires could extend from first conductive stem **410** through cavity **114** of tube body **12** and be connected to wires **452** at second conductive stem **412** of DWG **400** in the downstream sub-assembly **200**.

(160) In this embodiment, gun tube **10** is pressed into outer casing **700**. Outer casing **700** has a first end **702** with internal threads **702A**, a second end **704** with internal threads **704A**, an outer surface **706** and an internal cavity **708B** with an inner surface **708A**. When gun tube **10** is pressed into internal cavity **708B**, grounding hardware items **70**, which may be ball plungers, are compressed to their second compressed position, and they bias back to the first, extended position when they align with grooves (not shown) on inner surface **708A** that have a slightly larger diameter than the rest of internal cavity **708B**. In that manner, gun tube **10** is affixed in position in outer casing **700**.

(161) After gun tube **10** is positioned, sub-assemblies **200** are screwed onto each end **702**, **704** of outer casing **700**. As best seen in FIG. **26**, when assembled, second conductive stem structure **312** of plunger **300** in forward sub-assembly **200** is in contact with electrical contact **50** of first end fitting **16**. First conductive stem structure **310** of plunger **300** in rear sub-assembly **200** contacts electrical contact **50** of second end fitting **20**.

(162) Some non-limiting examples of embodiments of this disclosure follow:

Example Set 1

(163) Example 1: A plunger configured to fit in a central bore of a sub-assembly for a wellbore perforating gun assembly, the plunger comprising: an outer casing comprised of insulating material and having a first end; a first end portion comprised of electrically conductive material and

including a first conductive stem, the first conductive stem having a first, extended position, and a second, contracted position.

(164) Example 2: The plunger of example 1, wherein the outer casing further comprises a second end; and the plunger further comprises a second end portion comprised of electrically conductive material and including a second conductive stem, the second conductive stem having a first, extended position and a second, contracted position.

(165) Example 3: The plunger of example 1 or 2, wherein the distance between the first, extended position of the first conductive stem and the second, contracted position of the first conductive stem is from .150" to 1.250".

(166) Example 4: The plunger of example 2, wherein the difference between the first, extended position of the second conductive stem and the second, contracted position of the second conductive stem is from .150" to 1.250".

(167) Example 5: The plunger of example 1 or 4, wherein the distance between the first, extended position of the first conductive stem and the second, contracted position of the first conductive stem is from .150" to 1.250".

(168) Example 6: The plunger of any of examples 1-5, wherein the first end portion further includes a first cylinder connected to the first conductive stem and positioned inside of the outer housing, wherein the first cylinder has a diameter that is greater than a diameter of the first conductive stem.

(169) Example 7: The plunger of any of examples 2 or 4-6, wherein the second end portion further includes a second cylinder connected to the second conductive stem and positioned inside of the outer housing, wherein the second cylinder has a diameter that is greater than a diameter of the second conductive stem.

(170) Example 8: The plunger of any of examples 1-7, wherein the first conductive stem has a first distal tip that is positioned past the first end of the outer casing when the first conductive stem is in its first, extended position.

(171) Example 9: The plunger of any of examples 2 or 4-6, wherein the second conductive stem has a second distal tip that is positioned past the second end of the outer casing when the second conductive stem is in its first, extended position.

(172) Example 10: The plunger of any of examples 1-9 that further comprises a first spring that biases the first conductive stem to its first, extended position, wherein the spring is compressed when the first conductive stem is in its second, contracted position.

(173) Example 11: The plunger of any of examples 2, 4-6, or 9 that further comprises a second spring that biases the second conductive stem to its second, extended position, wherein the spring is compressed when the second conductive stem is in its second, contracted position.

(174) Example 12: The plunger of example 11 that further comprises a first spring that biases the first conductive stem to its first, extended position, wherein the spring is compressed when the first conductive stem is in its second, contracted position.

(175) Example 13: The plunger of example 12, wherein the first spring and the second spring each has a compressive force from 5 lbs. to 15 lbs.

(176) Example 14: The plunger of example 12, wherein the first spring and the second spring each has a compressive force from 2 lbs. to 20 lbs.

(177) Example 15: The plunger of example 12, wherein the first spring and the second spring each has a compressive force from 5 lbs. to 30 lbs.

(178) Example 16: The plunger of any of examples 1-15 that has an outer casing length of between 2" and 12".

(179) Example 17: The plunger of any of examples 1-16 that has an outer casing length of between 2" and 5".

(180) Example 18: The plunger of any of examples 1-17, wherein the insulating material is plastic.

(181) Example 19: The plunger of any of examples 1-18, wherein the outer casing has an outer

surface and at least one annular groove on the outer surface, and an O-ring in the at least one annular groove.

(182) Example 20: The plunger of any of examples 1-19 that has two annular grooves on the outer surface, and an O-ring in each of the two annular grooves.

(183) Example 21: The plunger of example 6, wherein the first cylinder is integrally formed with the first conductive stem.

(184) Example 22: The plunger of example 10 that further comprises a conductive inner core and the first end portion further includes a first cylinder, the first cylinder being positioned inside of the outer housing, and the first spring being positioned between the conductive inner core and the first cylinder.

(185) Example 23: The plunger of example 11 that further comprises a conductive inner core, and the second end portion further includes a second cylinder, the second cylinder being positioned inside of the outer housing, and the second spring being between the conductive inner core and the second cylinder.

(186) Example 24: The plunger of example 7, wherein the second cylinder is integrally formed with the second conductive stem.

(187) Example 25: The plunger of any of examples 1-24, wherein the first end is configured to be rotated by a tool.

(188) Example 26: The plunger of example 25, wherein the first end has a shape selected from the group consisting of one of the following: hexagonal, Torx, quadrangle, Allen head, Star drive, and other driving configuration.

(189) Example 27: A sub-assembly having a first end with a first opening, a second end with a second opening, and a central bore between the first opening and the second opening, and the plunger of example 2 positioned in the central bore and configured so the first, conductive stem is positioned at least partially in the first opening.

(190) Example 28: The sub-assembly of example 27, wherein the first opening has a surface, and the central bore has a surface, and that further includes a dart retainer that surrounds at least part of the first conductive stem and contacts the surface of the central bore.

(191) Example 29: The sub-assembly of example 28, wherein the dart retainer has a first section with a first diameter, a second section with a second diameter, and an opening therethrough, and the first conductive stem is positioned in the opening, and the first section contacts the surface of the central bore, and the second section contacts the surface of the first opening.

(192) Example 30: The sub-assembly of example 29, wherein the dart retainer is comprised of silicone rubber.

(193) Example 31: The sub-assembly of any of examples 27-30 that further comprises a second conductive stem having a second distal tip that is positioned outside of the central bore and positioned in the second opening.

(194) Example 32: The sub-assembly of any of examples 27-31, wherein the first conductive stem has a first distal tip that is positioned outside of the central bore and positioned outside of the first opening.

(195) Example 33: The sub-assembly of any of examples 27-32 that further comprises a second conductive stem having a distal tip that is positioned outside of the central bore and positioned outside of the second opening.

(196) Example 34: The sub-assembly of example 28, wherein the second conductive stem is positioned at least partially in the second opening, and that further includes a dart retainer that surrounds at least part of the first second conductive stem and contacts the surface of the central bore.

(197) Example 35: The sub-assembly of example 34, wherein the dart retainer has a first section with a first diameter, a second section with a second diameter, and an opening therethrough, and the second conductive stem is positioned in the opening, and the first section contacts the surface of the

central bore, and the second section contacts the surface of the second opening.

Example Set 2

(198) Example 1: A gun tube comprising: a body having a first end, a second end, a cavity, and a longitudinal axis; one or more weights in the cavity, the one or more weights configured to rotate the body around the longitudinal axis based on gravity acting on the one or more weights; and a first end fitting attached to the first end of the body, the first end fitting rotationally connected to the body.

(199) Example 2: The gun tube of example 1, wherein the first end fitting includes a first bearing housing.

(200) Example 3: The gun tube of example 1 or 2 that further includes a second end fitting attached to the second end of the body, the second end fitting rotationally connected to the body.

(201) Example 4: The gun tube of example 3, wherein the second end fitting includes a second bearing housing.

(202) Example 5: The gun tube of any of examples 1-4, wherein the first end fitting further comprises a first end contact having a first, extended position and a second, contracted position.

(203) Example 6: The gun tube of any of examples 3-4, wherein the second end fitting comprises a second end contact having a first, extended position and a second, contracted position.

(204) Example 7: The gun tube of any of examples 1-6, wherein the one or more weights comprises two separate weights, a first weight and a second weight.

(205) Example 8: The gun tube of example 7, wherein the first weight is juxtaposed the first end of the tube body, and the second weight is juxtaposed the second end of the tube body.

(206) Example 9: The gun tube of any of examples 1-8, wherein each of the one or more weights has a semi-cylindrical shape.

(207) Example 10: The gun tube of example 7, wherein the first weight weighs $\frac{7}{8}$ lbs. at sea level and the second weight weighs $1\frac{3}{4}$ lbs. at sea level.

(208) Example 11: The gun tube of example 7, wherein the second weight is at least twice as heavy as the first weight.

(209) Example 12: The gun tube of any of examples 1-11, wherein the one or more weights collectively weigh from 2 lbs. to 8 lbs. at sea level.

(210) Example 13: The gun tube of any of examples 1-12, wherein the one or more weights are comprised of steel.

(211) Example 14: The gun tube of any of examples 1-13, wherein the one or more weights is collectively one of the following percentages of the weight of the gun tube without the weight: at least 15%, at least 20%, at least 30%, at least 40%, and at least 50%.

(212) Example 15: The gun tube of example 7, wherein the first weight is 2"-3" in length and the second weight is 3"-8" in length.

(213) Example 16: The gun tube of any of examples 1-15, wherein the at least first end fitting comprises: an outer collar; a bearing housing that includes ball bearings and a central opening; and a support having a first portion with a first diameter and a second portion with a second diameter that is greater than the first diameter, wherein the bearing housing is positioned on the first portion and the central opening surrounds at least part of the first portion, and the outer collar is fastened to the support.

(214) Example 17: The gun tube of any of examples 1-16 that further comprises one or more charge openings configured to receive an explosive charge.

(215) Example 18: The gun tube of example 17 that further comprises one or more explosive charges in the one or more charge openings.

(216) Example 19: The gun tube of example 17 that further comprises one or more clip openings configured to receive charge clips.

(217) Example 20: The gun tube of example 19 that comprises one or more clips in the one or more clip openings.

(218) Example 21: The gun tube of example 16, wherein the first end fitting further includes a first end contact having a first, extended position and a second, contracted position, and that also comprises a second end fitting having a second end contact including a first, extended position and a second, extended position.

(219) Example 22: The gun tube of example 16, wherein the outer collar has one or more openings, wherein at least one of the one or more openings contains grounding hardware biased to a first, extended position, and that also has a second, contracted position.

(220) Example 23: The gun tube of any of examples 1-22, wherein the first end fitting comprises an end contact having a first end that comprises a stem, the stem being positioned inside of the cavity, and the end contact having a second end, the second end comprising an electrical contact that is positioned outside of the body.

(221) Example 24: The gun tube of example 23, wherein the end contact is configured to transmit electricity therethrough.

(222) Example 25: The gun tube of any of examples 1-24, wherein the first end fitting comprises a first end contact that includes a housing and one or more frangible elements extending outwardly from the housing.

(223) Example 26: The gun tube of example 25 that further comprises a second end fitting that includes a second end contact having a housing and one or more frangible elements extending outwardly from the housing.

(224) Example 27: The gun tube of example 25 or 26, wherein the housing and frangible elements are comprised of plastic and the frangible elements are configured to break away from the housing upon the application of explosive, outward axial force caused by explosion of one or more explosive charges in the gun tube.

(225) Example 28: The gun tube of example 5, wherein the first end contact is biased towards the first, extended position.

(226) Example 29: The gun tube of example 6, wherein the second end contact is biased towards the first, extended position.

(227) Example 30: The gun tube of example 28 that further includes a spring on a housing of the first end contact, the spring configured to bias the first end contact to the first, extended position, and the spring configured to compress when the first end contact moves to its second, contracted position.

(228) Example 31: The gun tube of example 29 that further includes a spring on a housing of the second end contact, the spring configured to bias the first end contact to the first, extended position, and the spring configured to compress when the first end contact moves to its second, contracted position.

(229) Example 32: The gun tube of example 5, wherein the end fitting includes an opening in which the first end contact is positioned.

(230) Example 33: The gun tube of any of examples 25-27, wherein the first end fitting further includes a support that has an opening configured to receive the one or more frangible elements, and wherein the first end contact has a first rotated position in which the one or more frangible elements fit through the opening and a second rotated position in which the one or more frangible elements do not fit through the opening.

(231) Example 34: The gun tube of example 27, wherein the one or more frangible elements are configured to break away from the housing when about 30 lbs. or more of explosive, outward longitudinal axial force is applied to them.

(232) Example 35: The gun tube of example 5, wherein the first end contact comprises a stem that includes a through hole, the through hole configured to receive one or more wires.

(233) Example 36: The gun tube of example 6, wherein the second end contact comprises a stem that includes a through hole, the through hole configured to receive one or more wires.

(234) Example 37: The gun tube of any of examples 1-36, wherein the body further comprises a

plurality of tabs for retaining the one or more weights.

(235) Example 38: The gun tube of any of examples 1-37 that further includes tabs at different positions on the body to maintain the one or more weights at different, respective positions within the cavity.

(236) Example 39: The gun tube of any of examples 1-38, wherein the body further comprises tabs that have a first, open position, and a second, closed position in which the tabs retain the one or more weights in the cavity.

(237) Example 40: The gun tube of any of examples 1-39 that further includes an outer casing positioned over and around the body, the outer casing having a first end and a second end.

(238) Example 41: The gun tube of example 39 that further comprises a sub-assembly connected to one end of the outer casing.

(239) Example 42: The gun tube of example 39 that further comprises a first sub-assembly connected to the first end of the outer casing and a second sub-assembly connected to the second end of the outer casing.

(240) Example 43: The gun tube of example 41, wherein the sub-assembly is threadingly connected to the outer casing.

(241) Example 44: The gun tube of example 42, wherein the first sub-assembly is threadingly connected to the first end of the outer casing and the second sub-assembly is threadingly connected to the second end of the outer casing.

(242) Example 45: The gun tube of example 41 that further comprises a plunger in the sub-assembly.

(243) Example 46: The gun tube of example 45, wherein the plunger has a longitudinal axis and an electrical connection running through it.

(244) Example 47: The gun tube of example 45 that further includes an electrically insulating outer casing around at least part of the plunger and the outer casing has a first end and a second end.

(245) Example 48: The gun tube of example 47, wherein the electrically insulating casing is comprised of plastic.

(246) Example 49: The gun tube of example 43, wherein the plunger has a body, a cavity, a first end, and a second end, a first conductive stem, and a second conductive stem, wherein the first contact stem extends past the first end of the outer casing, and the second contact stem extends past the second end of the outer casing.

(247) Example 50: The gun tube of example 49, wherein the first conductive stem has a first, extended position and a second, contracted position.

(248) Example 51: The gun tube of example 50, wherein the second conductive stem has a first, extended position and a second, contracted position.

(249) Example 52: The gun tube of example 50, wherein the distance between the first, extended position and the second, contracted position of the first conductive stem is between .150" and 1.250".

(250) Example 53: The gun tube of example 51, wherein the distance between the first, extended position and the second, contracted position of the second conductive stem is between .150" and 1.250".

(251) Example 54: The gun tube of example 50, wherein the first conductive stem is part of a first conductive stem structure that includes a first cylinder that is positioned in a cavity of the outer casing.

(252) Example 55: The gun tube of example 51, wherein the second conductive stem is part of a first conductive stem structure that includes a second cylinder that is positioned in a cavity of the outer casing.

(253) Example 56: The gun tube of example 54, wherein the cavity includes a conductive core, and a spring is positioned between the first conductive stem structure base and the conductive core.

(254) Example 57: The gun tube of example 56, wherein the cavity includes a conductive core, and

- a spring is positioned between the second conductive stem structure base and the conductive core.
- (255) Example 58: The gun tube of example 45, wherein the plunger has an outer casing and a compressible metal clip positioned on the outside surface, the metal clip configured to provide an electrical ground for the plunger.
- (256) Example 59: The gun tube of example 45, wherein there is a through hole in the first conductive stem.
- (257) Example 60: The gun tube of example 45, wherein there is a through hole in the second conductive stem.
- (258) Example 61: The gun assembly of example 45 or 51 that further includes an insulating barrel connector mounted to the second stem.
- (259) Example 62: The gun tube of example 45, wherein the plunger further comprises an outer casing and a driver head on a first end or a second end of the outer casing.
- (260) Example 63: The gun tube of example 16, wherein the collar includes one or more apertures, and each aperture includes a grounding mechanism to ground the gun tube when positioned inside of an outer casing.
- (261) Example 64: The gun tube of example 63, wherein each of the grounding mechanisms is a ball and plunger unit.
- (262) Example 65: The gun tube of example 63, wherein each grounding mechanism has a first, outwardly-biased position and a second, contracted position.
- (263) Example 66: The gun tube of example 65, wherein the distance between the first, outwardly-biased position and the second, contracted position from .010" to .080".
- (264) Example 67: The gun tube of example 1 that includes at least one rotatable end plate that is rotatable to a plurality of indexed positions, wherein the end plate is attached to one of the one or more weights.
- (265) Example 68: The gun tube of example 67 that includes one end plate at the first end of the gun tube.
- (266) Example 69: The gun tube of example 68 that includes a second rotatable end plate that is rotatable to a plurality of indexed positions, wherein the second end plate is attached to the one or more weights.
- (267) Example 70: The gun tube of example 69, wherein the first rotatable plate includes a plurality of indexed positions, and the second rotatable plate includes the same plurality of indexed positions.

Example Set 3

- (268) Example 1: A double-wire feed through with ground (DWG) comprising: an outer casing comprised of insulating material, the outer casing having a first end and a second end; a first conductive stem extending outward from the first end of the outer casing, the first conductive stem having a first, extended position and a second, contracted position.
- (269) Example 2: The DWG of example 1 that further comprises one or more grounding legs attached to and extending outward from the outer casing.
- (270) Example 3: The DWG of example 2 that includes two grounding legs, a first grounding leg and a second grounding leg.
- (271) Example 4: The DWG of example 3, wherein the first grounding leg is on one side of the outer casing and the second grounding leg is on the opposite side of the outer casing.
- (272) Example 5: The DWG of example 1 or 2, wherein the outer casing further comprises one or more recesses, and each of the one or more recesses is configured to receive a grounding leg when the grounding leg is compressed.
- (273) Example 6: The DWG of any of examples 1-5 that further includes a second conductive stem opposite the first conductive stem and an insulating sheath that connects one or more wires to the second conductive stem.
- (274) Example 7: The DWG of any of examples 1-6 that further includes a conductive core and a

spring between the conductive core and the first conductive stem, wherein the spring is configured to bias the first conductive stem to its first, extended position.

(275) Example 8: The DWG of example 7 that further includes a second conductive stem opposite the first conductive stem and an insulating sheath that connects one or more wires to the second conductive stem.

(276) Example 9: The DWG of any of examples 1-8, wherein the distance between the first, extended position and the second, contracted position is from .150" to 1.250".

(277) Example 10: The DWG of example 7, wherein the spring has a compressive force from 5 lbs. to 15 lbs.

(278) Example 11: The DWG of example 7, wherein the spring has a compressive force from 2 lbs. to 20 lbs.

(279) Example 12: The DWG of example 7, wherein the spring has a compressive force from 5 lbs. to 30 lbs.

(280) Example 13: A double-wire feed through with ground (DWG) comprising: an outer casing comprised of insulating material, the outer casing having a first end and a second end; a first conductive stem extending outward from the first end of the body, and a second conductive stem opposite the first conductive stem; and one or more grounding legs attached to and extending outward from the outer casing.

(281) Example 14: The DWG of example 13 that includes two grounding legs.

(282) Example 15: The DWG of example 13 that further includes an insulating sheath that connects one or more wires to the second conductive stem.

(283) Example 16: The DWG of example 1, wherein the insulating material comprises plastic.

(284) Example 17: The DWG of example 13, wherein the insulating material comprises plastic.

(285) Example 18: The DWG of example 2, wherein each of the one or more grounding legs extends outward from the outer casing by .050" to .250".

(286) Example 18: The DWG of example 13, wherein each of the one or more grounding legs extends outward from the outer casing by .050" to .250".

(287) Example 20: A sub-assembly having a first end with a first opening, a second end with a second opening, and a central bore between the first opening and the second opening, and the DWG of example 1 positioned in the central bore and configured so the first, conductive stem is positioned at least partially in the first opening.

(288) Example 21: The sub-assembly of example 20, wherein the first opening has a surface, and the central bore has a surface, and that further includes a dart retainer that surrounds at least part of the first conductive stem and that contacts the surface of the central bore.

(289) Example 22: The sub-assembly of example 21, wherein the dart retainer has a first section with a first diameter, a second section with a second diameter, and a retainer opening therethrough, and the first stem is positioned in the retainer opening, and the first section contacts the surface of the central bore, and the second section contacts the surface of the first opening.

(290) Example 23: The sub-assembly of example 21 or 22, wherein the dart retainer is comprised of silicone rubber.

(291) Example 24: A sub-assembly having a first end with a first opening, a second end with a second opening, and a central bore between the first opening and the second opening, and the DWG of example 13 positioned in the central bore and configured so the first, conductive stem is positioned at least partially in the first opening.

(292) Example 25: The sub-assembly of example 24, wherein the first opening has a surface, and the central bore has a surface, and that further includes a dart retainer that surrounds at least part of the first conductive stem and contacts the surface of the central bore.

(293) Example 26: The sub-assembly of example 25 or 26, wherein the dart retainer has a first section with a first diameter, a second section with a second diameter, and a retainer opening therethrough, and the first stem is positioned in the retainer opening, and the first section contacts

the surface of the central bore, and the second section contacts the surface of the first opening.
(294) Example 27: The sub-assembly of example 25, wherein the dart retainer is comprised of silicone rubber.

Example Set 4

(295) Example 1: An end fitting comprising: a first end and a second end; a bearing housing that includes ball bearings, the bearing housing having a bearing opening; a support having a first portion with a first diameter and a second portion with a second diameter that is greater than the first diameter, wherein the bearing housing is positioned on the first portion with the bearing opening surrounding at least part of the first portion; and an end contact comprising a housing, a first end having a conductive stem, and a second end that comprises an electrical contact, the second end having a first, extended position and a second, contracted position.

(296) Example 2: The end fitting of example 1, wherein the end contact is biased to the first, extended position.

(297) Example 3: The end fitting of example 1 or 2, wherein electricity can be conducted through the end contact.

(298) Example 4: The end fitting of any of examples 1-3, wherein the end contact further comprises a housing and one or more frangible elements extending outwardly from the housing.

(299) Example 5: The end fitting of example 4, wherein the housing and the one or more frangible elements are comprised of plastic.

(300) Example 6: The end fitting of example 4 or 5, wherein the one or more frangible elements are a plurality of tabs.

(301) Example 7: The end fitting of example 6, wherein the one or more frangible elements are two tabs.

(302) Example 8: The end fitting of example 6, wherein each of the plurality of tabs extend outward from the body by .070" to .125".

(303) Example 9: The end fitting of example 6, wherein each of the plurality of tabs is from .010" to .080" thick.

(304) Example 10: The end fitting of example 8, wherein each of the plurality of tabs is from .010" to .080" thick.

(305) Example 11: The end fitting of example 2 that further includes a spring on the end contact.

(306) Example 12: The end fitting of example 11, wherein the spring is on a first portion of the end contact.

(307) Example 13: The end fitting of example 12, wherein the support further includes one or more frangible elements and the spring is retained between a central portion of the end contact and the one or more frangible elements.

(308) Example 14: The end fitting of example 6, wherein the support has an opening that receives an end of the end contact housing that includes the plurality of tabs, and wherein the end contact has a first position in which the tabs fit through the opening and a second position in which they do not fit through the opening.

(309) Example 15: The end fitting of example 4, wherein the one or more frangible elements break when 30 lbs. or more of explosive, outward, longitudinal, axial force is applied to them.

(310) Example 16: The end fitting of example 4, wherein the one or more frangible elements break when 50 lbs. or more of explosive, outward, axial force is applied to them.

(311) Example 17: The end fitting of any of examples 1-16, wherein the conductive stem includes a through hole, wherein the through hole is configured to receive one or more wires.

(312) Example 18: The end fitting of any of examples 1-17 that further includes a wire harness assembly attached to the conductive stem, the wire harness assembly comprising an insulated wire and an insulated circular connector.

(313) Example 19: The end fitting of example 18, wherein the insulated circular connector is a barrel crimp connector.

- (314) Example 20: An end fitting for a gun tube that comprises an end contact with a first end that includes an electrical contact having a first extended position and a second, contracted position.
- (315) Example 21: The end fitting of example 20, wherein the end contact further includes one or more frangible elements configured to break when 30 lbs. or more of explosive, outward longitudinal, axial, force is applied.
- (316) Example 22: The end fitting of example 21, wherein the one or more frangible elements are a plurality of tabs.
- (317) Example 23: The end fitting of example 22, wherein the one or more frangible elements are two tabs.
- (318) Example 24: The end fitting of any of examples 1-23 that further comprises an outer collar having an opening therethrough.
- (319) Example 25: The end fitting of example 24, wherein the electrical contact is positioned from 1/16" to 5/16" outside of the opening when the second end of the end contact is in its first, extended position.
- (320) Example 26: The end fitting of example 4, wherein the housing and one or more frangible elements are integrally formed.

Example Set 5

- (321) Example 1: A gun tube comprising: a body having a cavity, a longitudinal axis, a first end, and a second end; a motor connected to the first end, the motor configured to rotate the body around the longitudinal axis.
- (322) Example 2: The gun tube of example 1 that further comprises a first end fitting attached to the first end of the body.
- (323) Example 3: The gun tube of example 2 that further comprises a second end fitting attached to the second end of the body.
- (324) Example 4: The gun tube of example 1 that further comprises a sensor configured to detect the location of the explosive charges.
- (325) Example 5: The gun tube of example 3, wherein the sensor comprises an accelerometer.
- (326) Example 6: The gun tube of example 3, wherein the sensor comprises one or more of an accelerometer, a magnetometer, and gyroscope.
- (327) Example 7: A system comprising the gun tube of example 6 and a motor control remote to the gun tube, the motor control configured to operate the motor.
- (328) Example 8: The system of example 7, wherein the motor control is one of a computer and a cell phone.
- (329) Example 9: The system of example 7 that further includes a receiver for receiving transmissions sent by the sensor.
- (330) Example 10: The system of a claim 7, wherein the motor control is configured to be operated by a human operator.
- (331) Example 11: The system of a claim 7, wherein the motor control is configured to be operated by a machine operator.
- (332) Example 12: The gun tube of example 1, wherein the at least first end fitting comprises: an outer collar; a bearing housing that includes ball bearings and a central opening; and a support having a first portion with a first diameter and a second portion with a second diameter that is greater than the first diameter, wherein the bearing housing is positioned on the first portion and the central opening surrounds at least part of the first portion, and the outer collar is fastened to the support.
- (333) Example 13: The gun tube of any of examples 1-12 that further comprises one or more charge openings configured to receive an explosive charge.
- (334) Example 14: The gun tube of example 13 that further comprises one or more explosive charges in the one or more charge openings.
- (335) Example 15: The gun tube of any of examples 1-14 that further comprises one or more clip

openings configured to receive charge clips.

(336) Example 16: The gun tube of example 15 that comprises one or more clips in the one or more clip openings.

(337) Example 17: The gun tube of example 2, wherein the first end fitting includes a first end contact having a first, extended position and a second, contracted position, and that also comprises a second end fitting having a second end contact including a first, extended position and a second, extended position.

(338) Example 18: The gun tube of example 12, wherein the outer collar has one or more openings, wherein at least one of the one or more openings contains grounding hardware biased to a first, extended position, and that also has a second, contracted position.

(339) Example 19: The gun tube of example 2 or 17, wherein the first end fitting comprises an end contact having a first end that comprises a stem, the stem being positioned inside of the cavity, and the end contact having a second end, the second end comprising an electrical contact that is positioned outside of the body.

(340) Example 20: The gun tube of example 19, wherein the end contact is configured to transmit electricity therethrough.

(341) Example 21: The gun tube of example 2, wherein the first end fitting comprises a first end contact that includes a housing and one or more frangible elements extending outwardly from the housing.

(342) Example 22: The gun tube of example 21 that further comprises a second end fitting that includes a second end contact having a housing and one or more frangible elements extending outwardly from the housing.

(343) Example 23: The gun tube of example 21, wherein the housing and frangible elements are comprised of plastic and the frangible elements are configured to break away from the housing upon the application of explosive, outward axial force caused by explosion of one or more explosive charges in the gun tube.

(344) Example 24: The gun tube of example 17, wherein the first end contact is biased towards the first, extended position.

(345) Example 25: The gun tube of example 24, wherein the second end contact is biased towards the first, extended position.

(346) Example 26: The gun tube of example 24 that further includes a spring on a housing of the first end contact, the spring configured to bias the first end contact to the first, extended position, and the spring configured to compress when the first end contact moves to its second, contracted position.

(347) Example 27: The gun tube of example 26 that further includes a spring on a housing of the second end contact, the spring configured to bias the first end contact to the first, extended position, and the spring configured to compress when the first end contact moves to its second, contracted position.

(348) Example 28: The gun tube of example 17, wherein the distance between the first, extended position and the second, contracted position of the first end contact is between .150" and 1.250".

(349) Example 29: The gun tube of example 28, wherein the distance between the first, extended position and the second, contracted position of the second end contact is between .150" and 1.250".

(350) Having thus described different embodiments, other variations and embodiments that do not depart from the spirit of this disclosure will become apparent to those skilled in the art. The scope of the claims is thus not limited to any particular embodiment, but is instead set forth in the claims and the legal equivalents thereof. Unless expressly stated in the written description or claims, the steps of any method recited in the claims may be performed in any order capable of yielding the desired product. No language in the specification should be construed as indicating that any non-claimed limitation is included in a claim. The terms "a" and "an" in the context of the following claims are to be construed to cover both the singular and the plural, unless otherwise indicated.

Claims

1. A downhole perforating gun system, comprising: an outer body configured to connect to the downhole perforating gun system; an inner body disposed inside the outer body and having at least one explosive charge receptacle, the inner body having a longitudinal axis and having a center of gravity offset from the longitudinal axis of the inner body; a first rotational assembly comprising a first structure and a second structure, the first and second structures being rotatable relative to one another, the first structure engaged with at least a portion the inner body, the second structure engaged with at least a portion of the outer body, the first rotational assembly rotatably supporting the inner body relative to the outer body and being configured to permit radial orientation of the inner body about the longitudinal axis relative to the outer body based on the center of gravity of the inner body; and a first electrical conductor at least partially disposed through the first rotational assembly and being configured to conduct electricity with the downhole perforating gun system.
2. The downhole perforating gun system of claim 1, wherein the first structure of the first rotational assembly comprises a first housing member of a bearing housing; wherein the second structure of the first rotational assembly comprises a second housing member of the bearing housing; and wherein the first rotational assembly comprises one or more bearings disposed between the first and second housing members of the bearing housing.
3. The downhole perforating gun system of claim 1, wherein the first electrical conductor comprises a first electrical contact being associated with the inner body and being biased between an extended position and a contracted position.
4. The downhole perforating gun system of claim 1, wherein the outer body comprises: an outer casing having first and second ends and defining an interior therein, the inner body disposed inside the interior of the outer casing; and a first sub-assembly configured to connect to the first end of the outer casing, wherein the second structure of the first rotational assembly is engaged with at least a portion of the first sub-assembly.
5. The downhole perforating gun system of claim 1, wherein the outer body comprises: an outer casing having first and second ends and defining an interior therein, the inner body disposed inside the interior of the outer casing; and a first sub-assembly configured to connect to the first end of the outer casing, wherein the second structure of the first rotational assembly is engaged with at least a portion of the outer casing.
6. The downhole perforating gun system of claim 5, wherein the second structure of the first rotational assembly is further engaged with at least a portion of the first sub-assembly.
7. The downhole perforating gun system of claim 1, wherein the first electrical conductor comprises a first electrical contact associated with the inner body; and wherein the outer body comprises: an outer casing having first and second ends and defining an interior therein, the inner body disposed inside the interior of the outer casing; and a first sub-assembly configured to connect to the first end of the outer casing, the first sub-assembly having a second electrical contact of the first electrical conductor, the second electrical contact being configured to electrically contact the first electrical contact.
8. The downhole perforating gun system of claim 7, wherein at least a portion of the first electrical contact is biased relative to the second electrical contact.
9. The downhole perforating gun system of claim 7, wherein at least a portion of the second electrical contact is biased relative to the first electrical contact.
10. The downhole perforating gun system of claim 1, wherein the inner body comprises one or more weight elements disposed on the inner body and providing the inner body with the center of gravity offset from the longitudinal axis of the inner body.
11. The downhole perforating gun system of claim 10, wherein the one or more weight elements are disposed at one or more predefined locations on the inner body and are retained by one or more

tabs to the inner body.

12. The downhole perforating gun system of claim 1, wherein the first rotational assembly rotatably supports a first end of the inner body; and the downhole perforating gun system further comprises: a second rotational assembly rotatably supporting a second end of the inner body relative to the outer body; and a second electrical conductor at least partially disposed through the second rotational assembly and being configured to conduct electricity with the downhole perforating gun system.

13. A downhole perforating gun system, comprising: an outer body comprising an outer casing and a first sub-assembly, the outer casing having first and second ends and defining an interior therein, the first sub-assembly being configured to connect the first end of the outer casing to the downhole perforating gun system; an inner body disposed in the interior of the outer casing and having at least one explosive charge receptacle, the inner body having a longitudinal axis and having a center of gravity offset from the longitudinal axis of the inner body; a first rotational assembly comprising a first structure and a second structure, the first and second structures being rotatable relative to one another, the first structure engaged with at least a portion of the inner body, the second structure engaged with at least a portion of the first sub-assembly, the first rotational assembly rotatably supporting the inner body relative to the outer body and being configured to permit radial orientation of the inner body about the longitudinal axis relative to the outer body based on the center of gravity of the inner body; and a first electrical conductor at least partially disposed through the first rotational assembly, the first electrical conductor comprising a first electrical contact associated with the inner body and comprising a second electrical contact associated with the first sub-assembly, the first and second electrical contacts being configured contact one another and being configured to conduct electricity therebetween, at least one of the first and second electrical contacts being biased relative to the other one of the first and second electrical contacts.

14. The downhole perforating gun system of claim 13, wherein the first structure of the first rotational assembly comprises a first housing member of a bearing housing; wherein the second structure of the first rotational assembly comprises a second housing member of the bearing housing; and wherein the first rotational assembly comprises one or more bearings disposed between the first and second housing members of the bearing housing.

15. The downhole perforating gun system of claim 13, wherein the second structure of the first rotational assembly is engaged with at least a portion of the outer casing.

16. The downhole perforating gun system of claim 13, wherein at least a portion of the first electrical contact is biased along the longitudinal axis by a spring relative to the second electrical contact.

17. The downhole perforating gun system of claim 13, wherein at least a portion of the second electrical contact is biased along the longitudinal axis by a spring relative to the first electrical contact.

18. The downhole perforating gun system of claim 13, wherein the inner body comprises one or more weight elements disposed on the inner body and providing the inner body with the center of gravity offset from the longitudinal axis of the inner body.

19. The downhole perforating gun system of claim 18, wherein the one or more weight elements are disposed at one or more predefined locations on the inner body and being retained by one or more tabs to the inner body.

20. The downhole perforating gun system of claim 13, wherein at least a portion of the first electrical contact is insulated.

21. The downhole perforating gun system of claim 13, wherein the first rotational assembly rotatably supports the first end of the inner body; and the downhole perforating gun system further comprises: a second rotational assembly rotatably supporting the second end of the inner body; and a second electrical conductor at least partially disposed through the second rotational assembly and being configured to conduct electricity with the downhole perforating gun system.

22. A downhole perforating gun system, comprising: an outer body comprising an outer casing and a first sub-assembly, the outer casing having first and second ends and defining an interior therein, the first sub-assembly being configured to connect the first end of the outer casing to the downhole perforating gun system; an inner body disposed in the interior of the outer casing, the inner body having a longitudinal axis and having at least one explosive charge receptacle; one or more weight elements disposed on the inner body and providing the inner body with a center of gravity offset from the longitudinal axis of the inner body; a first rotational assembly comprising a first structure and a second structure, the first and second structures being rotatable relative to one another, the first structure engaged with at least a portion of the inner body, the second structure engaged with at least a portion of the first sub-assembly, the first rotational assembly rotatably supporting the inner body relative to the outer body and being configured to permit radial orientation of the inner body about the longitudinal axis relative to the outer body based on the center of gravity provided by the one or more weight elements; and a first electrical conductor at least partially disposed through the first rotational assembly, the first electrical conductor comprising a first electrical contact associated with the inner body and comprising a second electrical contact associated with the first sub-assembly, the first and second electrical contacts being configured to contact one another and being configured to conduct electricity therebetween, at least one of the first and second electrical contacts being biased relative to the other one of the first and second electrical contacts.
